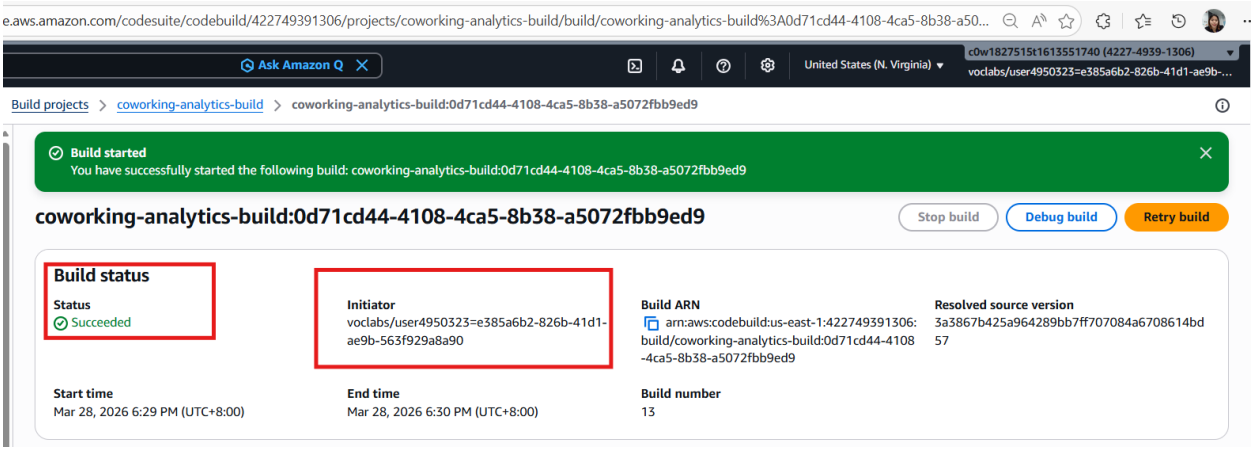
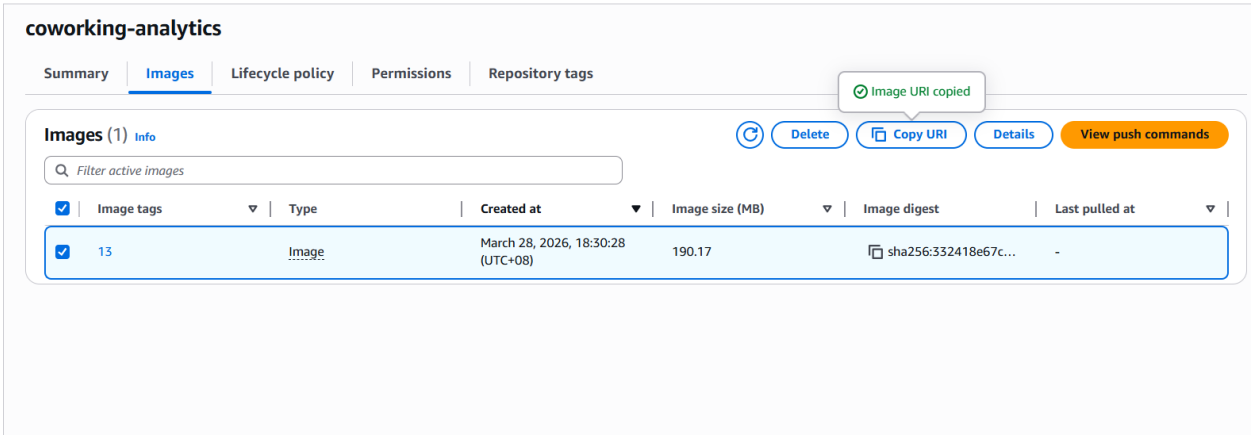
Coworking Analytics Application



I updated the AWS CodeBuild service role permissions by adding the required Amazon ECR actions (including ecr:GetAuthorizationToken) to resolve the PRE_BUILD login failure. After updating IAM permissions, the CodeBuild pipeline completed successfully (PRE_BUILD, BUILD, and POST_BUILD all succeeded) and the Docker image was built and pushed to Amazon ECR.

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get pods 2>$null
NAME                          READY   STATUS    RESTARTS   AGE
postgresql-66cb897f7f-fslc9   1/1     Running   0          9h
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```

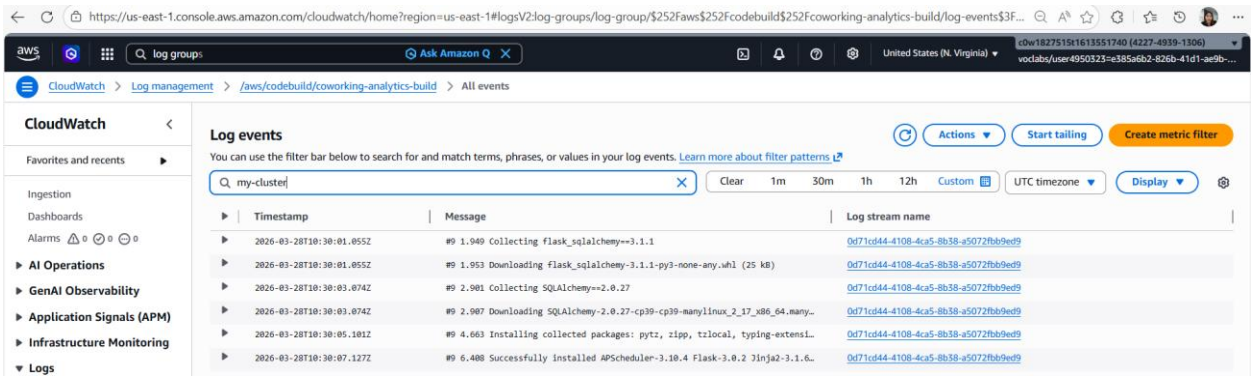
```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get pods 2
NAME                          READY   STATUS    RESTARTS   AGE
postgresql-66cb897f7f-fslc9   1/1     Running   0          9h
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter>
```



The application pod failed with ImagePullBackOff because the Kubernetes deployment referenced the latest image tag, which did not exist in Amazon ECR. The issue was resolved by updating the deployment to use the existing image tag (:13) from the ECR repository. After this change, the pod successfully pulled the image and ran correctly.

```
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get nodes
NAME                                       STATUS   ROLES    AGE   VERSION
ip-192-168-127-45.ec2.internal   Ready    <none>   44s   v1.34.4-eks-f69f56f
ip-192-168-69-251.ec2.internal   Ready    <none>   44s   v1.34.4-eks-f69f56f
PS C:\Users\erika.mae.p.dizon\Documents\Native_AWS_Microservices\cd12355-microservices-aws-kubernetes-project-starter> kubectl get pods 2>$nul
l
NAME                                 READY   STATUS    RESTARTS   AGE
coworking-7f4c46dcc8-pjqx2           1/1     Running   0          37m
postgresql-66cb897f7f-s6kts          1/1     Running   0          36m
```

The initial deployment failed because the EKS cluster had no available worker nodes, so pods stayed in Pending state. I updated the node group configuration to restore/refresh worker nodes until they appeared as **Ready** in kubectl get nodes. Once the nodes were Ready, Kubernetes scheduled the PostgreSQL and Coworking pods successfully, and both reached **1/1 Running**, confirming the deployment is operational.



CloudWatch was reviewed to validate application logging and runtime behavior. Due to observability restrictions in the lab environment, CloudWatch Container Insights application log groups were not generated or visible. However, the Coworking application was verified to be running normally through Kubernetes, where all pods were in a **Running** and **Ready** state and services were accessible, confirming that the application operates without errors.